Listing 2.1 Extracting the meter reading from a utility bill using OCR to digitize consumption data

```
from PIL import Image
import pytesseract, re

def extract_meter_reading(text):
  pattern = r"Current Meter Reading:\s*(\d+)" #1
  match = re.search(pattern, text)
  return int(match.group(1)) if match else None

def extract_date(text):
  pattern = r"(\d{1,2}/\d{1,2}/\d{4})" #2
  matches = re.findall(pattern, text)
  return matches[-1] if matches else None

def read_bill():
  image = Image.open(r"./images/gasBill.png")
  text = pytesseract.image_to_string(image)

  meter_reading = extract_meter_reading(text)
  date = extract_date(text)

  return {
    "meter_reading": meter_reading,
    "date": date
  }
```

#1 A regular expression to extract the meter reading based on the format of the bill.
#2 There are multiple dates in this bill. I want the last date that appears.

Once I have configured the regular expression in the code to extract the data I require (date and meter reading), I can then run this code against the decade worth of PDF bills I have been emailed by my utility providers. The output of this is shown in figure 2.2, which captures my household energy consumption for the past 10 years. I do the same with my water bills so that I have a baseline of consumption before I build my digital twin.